

Module 9: Sampling Distributions

Background

- Population Vs Sample

- **Population:** The population refers to the entire group or collection of individuals, objects, or events that we are interested in learning or making inferences about.
- **Sample:** A sample refers to a subset of individuals, objects, or events that are selected from a larger population to provide information and make inferences about the entire population.
- **Example:** A survey is administered to undergraduate students at a college in Nebraska about their attitudes on the new learning management system used in all courses. Of the 40,322 students enrolled at the college, a random sample of 2,500 students were surveyed.
 - Population? 40, 322 students enrolled at the college
 - Sample? 2,500 students surveyed

- Solve equations using square roots → square both sides!

Examples:

- $\sqrt{x} = \frac{1}{4}$
 - $6 = \sqrt{\frac{9}{x}}$
 - $\frac{12}{x} = 3$
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Inference Basics

- **Statistic Vs Parameter**

- A **statistic** describes a characteristic of a **sample**.
- A **parameter** describes a characteristic of a **population**.

Example: The student body of composition of Northern Arizona University as of Fall 2022 consisted of 25% Hispanic students. In a statistics class of 30 students, each student was asked to randomly select a sample of 30 students within the university. Each student then calculated the proportion of hispanic students in their sample.

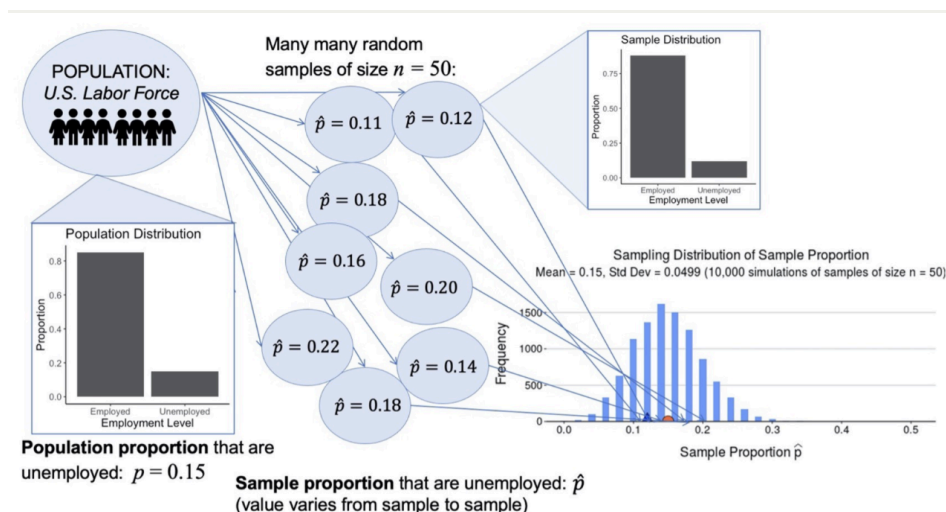
- Parameter? 25% proportion of Hispanic students at the university
- Statistic? Each student's calculated proportion of Hispanic students because it is derived from a sample, not the whole population.

Sampling Distribution of a Statistic:

- A probability distribution that describes the long-term behavior of a statistic from a large number of samples of size n from a given population.

Example: Back to the Northern Arizona University scenario... You draw a random sample of 50 students and find 11 students considered themselves Hispanic

- Population proportion, $p = 25\%$ or 0.25
- Sample proportion, $\hat{p} = \frac{11}{50} = 0.22$
- In order to get a sense of the pattern of variation in sample proportions, we need to generate more than five samples.
- Shows how sample proportions vary from sample to sample



Mean and Standard Deviation of Sampling Distribution for a Sample Proportion

- Sampling every possible sample is simply not feasible due to time, cost, and logistical constraints, especially when dealing with large populations. So, we use theoretical expressions for the mean and standard deviation.
 - Mean: $\mu_{\hat{p}} = p$
 - Standard Deviation: $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$

Example: Let's assume that the true unemployment rate (the proportion of the U.S. labor force that is unemployed) is 0.15. Calculate the mean and standard deviation of the sampling distribution of sample proportions using a sample size of 50. Round to 4 decimal places.

- $\mu_{\hat{p}} = 0.15$
- $\sigma_{\hat{p}} = \sqrt{\frac{0.15(1-0.15)}{50}} = \sqrt{0.00255} = 0.0505$

- **Importance of sample size:** In our example, the normality requirement may not have been satisfied due to a relatively **small sample size**.
 - Keep in mind that a **sufficiently large sample size** is key for the sampling distribution to approximate a normal distribution, allowing us to leverage powerful statistical tools confidently in our analyses.
- In practice we do not typically know the population proportion, nor can you typically take thousands of random samples. Instead, we observe a **single random sample**. In this case, we need to **estimate** the mean and standard deviation of the sample proportion.

- Estimated mean of population proportion: $\hat{p}$
- Estimated standard deviation: $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Example: Suppose you select a random sample of 100 individuals from the U.S. labor force and find that 12 of them are unemployed.

- What is the unemployment rate for this sample?
- Calculate the standard error of the sample proportion that are unemployed. Round to 4 decimal places.
- Write a sentence interpreting this value in context.

We estimate that the typical proportion of unemployed individuals in a random sample of 100 individuals from the U.S. labor force will fall about 0.0325 away from the proportion of unemployed individuals in the entire U.S. labor force

Sampling Variability

The center, spread, and shape of a sampling distribution of a sample proportion varies depending on the sample size, n , and the population proportion, p .

Exploration Activity: how sample proportions change for varying sizes of n .

- As the sample size increases, the center of the sampling distribution of sample proportions **remains the same**.
- As the sample size increases, the variability in the sampling distribution of sample proportions **decreases**.
- As the sample size increases, the shape of the sampling distribution of sample proportions **becomes more normal**.
- **Central Limit Theorem:** As the sample size gets **larger**, the distribution of the sample proportion will become closer to a **normal distribution**.
- **Normal Condition for Proportions:**

The Normal Condition for Proportions

Given a population proportion, p and a sample size n , if $np \geq 10$ and $n(1 - p) \geq 10$ then we assume that the sampling distribution of the sample proportion is approximately normal.

- The distribution of sample proportions appears normal when the sample size is sufficiently large

Example: A public health researcher is studying the proportion of individuals in a large city who are willing to participate in a new health program. Preliminary studies suggest that approximately 5% of the city's population is interested. The researcher wants to ensure that the sampling distribution of the sample proportion, $\hat{p}$, is approximately normal. Which of the following sample sizes would allow the sampling distribution to be considered approximately normal? (Select all that apply.)

- ☐ 30
- ☐ 100
- ☐ 250
- ☐ 500

Calculating Probabilities for the Sampling Distribution of the Sample Proportion

Example: A university reports that 65% of its current students are in-state residents. Let p denote the proportion of all current students at this university who are in-state residents: $p = 0.65$

In the simulated sampling distribution, it was found that the proportion of all current students at this university who are in-state residents was bell shaped. Rather than using a simulation, we can model the sampling distribution of sample proportions for samples of size 70 as a normal distribution.

- a) Find the mean.
- b) Find the standard deviation.
- c) Approximate the probability that you observe more than 50 in-state residents in a random sample of 70 students.

Example: Suppose it is known that 43% of Americans own an iPhone. If a random sample of 50 Americans were surveyed, what is the probability that the proportion of the sample who owned an iPhone is between 45% and 50%?

Calculate the Sample Size

If researchers want to limit their standard deviation, or variability, then they use the standard error formula to find the sample size.

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

Example: The university would like to do a follow-up study but would like the standard deviation of the sample proportion to be no higher than 0.01. Assume again that the true proportion of the current students who are in-state residents is 0.65. How many individuals would they need to ensure that the standard deviation of sample proportions is at most 0.01?